STATISTICS

10/11/21

CURVE FITTING:

This is a method of finding a specific relation connecting the dependent and independent variables for a given data so as to satisfy the data as accurately as possible. Such a curve is called the curve of best fit.

Least square method:

Curve fitting is a method of finding a suitable relation in the form $y = f(x)$ for a set of observed values (x_i, y_i) for $1 \le i \le n$. This relation is most suitable for predicting or estimating the value of y for a given value of x.

Suppose $y = f(x)$ is an approximate relation that fits into a given data comprising (x_i, y_i) for $1 \le i \le n$, then y_i's are called observed values and Y_i's are called expected values. $[Y_i = f(x_i)]$ $(y_1, y_2, y_3, \ldots)$ their difference $R_i = y_i - Y_i$ are called the residuals.

The method of least squares provides a relationship, $y = f(x)$ such that the sum of squares of the residuals is least.

1) Fitting of a straight line $y = ax + b$:

consider a set of 'n' given values (x, y) for fitting a straight line $y = ax + b$ where a & b are the parameters to be determined. The residual $R = y - (ax + b)$ is the difference b/n observed & estimated values of y. By the method of least squares, we find a and b such that the sum of squares of the residuals is minimum

1

(i.e, ΣR^2 is minimum.)

Let $P = \Sigma R^2$

i.e, $P = \Sigma (y - ax - b)^2$

Letting P as a function of a and b, the necessary conditions for P to be minimum are

$$\frac{\partial P}{\partial a} = 0 \quad \text{and} \quad \frac{\partial P}{\partial b} = 0$$

$\Rightarrow \Sigma 2(y - ax - b)(-x) = 0$

$\& \Sigma 2(y - ax - b)(-1) = 0$

$\Sigma (y - ax - b)(-x) = 0$ $\qquad$ $\Sigma (y - ax - b)(-1) = 0$

$-\Sigma xy + \Sigma ax^2 + \Sigma bx = 0$; $-\Sigma y + \Sigma ax + \Sigma b = 0$

$\therefore \Sigma y = a\Sigma x + nb$

$\Sigma xy = a\Sigma x^2 + b\Sigma x$

These eqns are called normal equations for fitting the straight line $y = ax + b$ in the least square ~~science~~ sense.

By solving these, we obtain values of a and b.

Note ① The normal equations for fitting the second degree parabola $y = ax^2 + bx + c$ are

$$\Sigma y = a\Sigma x^2 + b\Sigma x + nc$$

$$\Sigma xy = a\Sigma x^3 + b\Sigma x^2 + c\Sigma x$$

$$\Sigma x^2 y = a\Sigma x^4 + b\Sigma x^3 + c\Sigma x^2$$

② The normal eqns for fitting the exponential curve $y = ae^{bx}$ are

2

$y = ae^{bx}$

Taking log on both the sides,

$\log y = \log(ab^{bx})$

$\log y = \log a + bx \log e$

$\log y = \log a + bx$ $\quad Y = \log y \quad B = b$

$Y = A + BX$ where $\quad A = \log a \quad X = x$

which is a straight line whose normal equations are,

$\Sigma Y = B\Sigma X + nA$

& $\Sigma XY = B\Sigma x^2 + A\Sigma X$

Pblms:

1) Fit a straight line $y = ax + b$ for the following data:

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

Here $n = 8$

The normal eqns to fit a straight line $y = ax + b$ are

$\Sigma y = a\Sigma x + nb$

$\Sigma xy = a\Sigma x^2 + b\Sigma x$

x	y	xy	x^2
1	1	1	1
3	2	6	9
4	4	16	16
6	4	24	36
8	5	40	64
9	7	63	81
11	8	88	121
14	9	126	196
$\Sigma x = 56$	$\Sigma y = 40$	$\Sigma xy = 364$	$\Sigma x^2 = 524$

$56a + 8b = 40$

$524a + 56b = 364$

$a = 0.636 \simeq 0.64$

$b = 0.545 \simeq 0.55$

Thus required straight line is

$y = 0.64x + 0.55$

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2) Find the eqn of the best fitting straight line for the following data & hence estimate the value of y at $x = 30$.

x	5	10	15	20	25
y	16	19	23	26	30

Soln:- Here $n = 5$; Let $y = ax + b$ be the required straight line.

The normal equations to fit the straight line are

$$\Sigma y = a\Sigma x + nb$$

$$\Sigma xy = a\Sigma x^2 + b\Sigma x$$

x	y	xy	x^2
5	16	80	25
10	19	190	100
15	23	345	225
20	26	520	400
25	30	750	625
$\Sigma x = 75$	114	1885	1375

$$75a + 5b = 114$$

$$1375a + 75b = 1885$$

$\therefore a = 0.7$, $b = 12.3$

$\therefore$ Required straight line is $\boxed{y = 0.7x + 12.3}$

At $x = 30$
$y = 33.3$

3)

x	1	2	3	4	5
y	14	13	9	5	2

4)

x	0	1	2	3	4	5	6
y	2	1	3	2	4	3	5

1) Fit a second degree parabola in the formula $y = ax^2 + bx + c$ for the following data:

Hence estimate y at $x = 6$.

x	1	2	3	4	5
y	10	12	13	16	19

Here $n = 5$

Normal eqns to fit $y = ax^2 + bx + c$ are

$$\Sigma y = a\Sigma x^2 + b\Sigma x + nc$$

$$\Sigma xy = a\Sigma x^3 + b\Sigma x^2 + c\Sigma x$$

$$\Sigma x^2 y = a\Sigma x^4 + b\Sigma x^3 + c\Sigma x^2$$

	x	y	xy	x^2	x^3	x^2y	x^4
	1	10	10	1	1	10	1
	2	12	24	4	8	48	16
	3	13	39	9	27	117	81
	4	16	64	16	64	256	256
	5	19	95	25	125	475	625
Σ	15	70	232	55	225	906	979

$55a + 15b + 5c = 70$

$225a + 55b + 15c = 232$

~~$225a$~~ $979a + 225b + 55c = 906$

$\therefore$ $a = 0.285$ $b = 0.485$ $c = 9.4$

$a \simeq 0.29$ $\simeq 0.49$

$\therefore$ Required parabola $\Rightarrow y = 0.29x^2 + 0.49x + 9.4$

when $x = 6$, $y = 0.29(36) + 0.49(6) + 9.4$

$= 10.44 + 2.94 + 9.4$

$= \underline{\underline{22.78}}$

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2)

x	0	1	2	3	4
y	1	1.8	1.3	2.5	2.3

3)

x	0	1	2	3	4	5	6
y	14	18	23	29	36	40	46

4)

x	1	2	3	4	5	6	7	8	9
y	2	6	7	8	10	11	11	10	9

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2) 2) soln: Here $n=5$

Given eqn of parabola $y = ax^2 + bx + c$

Normal eqns are $\Sigma y = a\Sigma x^2 + b\Sigma x + nc$

$\Sigma xy = a\Sigma x^3 + b\Sigma x^2 + c\Sigma x$

$\Sigma x^2 y = a\Sigma x^4 + b\Sigma x^3 + c\Sigma x^2$

	x	y	x^2	x^3	x^4	xy	x^2y
	0	1	0	0	0	0	0
	1	1.8	1	1	1	1.8	1.8
	2	1.3	4	8	16	2.6	5.2
	3	2.5	9	27	81	7.5	22.5
	4	2.3	16	64	256	9.2	36.8
Σ	10	8.9	30	100	354	21.1	66.3

$30a + 10b + 5c = 8.9$

$100a + 354\ 30b + 10c = 21.1$

$354a + 100b + 30c = 66.3$

$\therefore a = -0.02$

$b = 0.415$

$c = 1.07$

$\therefore$ Required parabola is $y = -0.02x^2 + 0.4x + 1.07$

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1) Fit an exponential curve in the form $y = ae^{bx}$ for the following data

x	0	2	4
y	8.12	10	31.82

$Y = A + BX$

$Y = \log_e y \quad A = \log_e a$

	x	y	$\ln y$	x^2	$x \ln y$
	0	8.12	2.094	0	0
	2	10	2.302	4	4.604
	4	31.82	3.46	16	13.84
Σ	6		7.856	20	18.444

Normal eqns are

$\Sigma Y = B\Sigma x + nA$

$\Sigma xY = B\Sigma x^2 + A\Sigma x$

$n=3$

6

$\therefore$ $7.856 = 6B + 3A$

$18.444 = 20B + 6A$

$A = \ln a = 1.9356 \Rightarrow a = 6.928$

$B = 0.3415$

$\therefore$ Required curve is $y = 6.928e^{0.342x}$

2)

x	5	6	7	8	9	10
y	133	55	23	7	2	2

$Y = A + Bx$ where $Y = \log_e y$ and $A = \log_e a$

x	y	$\ln y$	x^2	$x \ln y$
5	133	4.89	25	24.45
6	55	4.007	36	~~330~~ 24.042
7	23	3.135	49	21.945
8	7	1.946	64	15.56
9	2	0.693	81	6.237
10	2	0.693	100	6.93
Σ 45		15.363	355	99.164 99.181

$\Sigma Y = B\Sigma x + nA$

$\Sigma xY = B\Sigma x^2 + A\Sigma x$

$6A + 45B = 15.364$

$45A + 355B = 99.181$

$A = 9.439 = \ln a \Rightarrow a = 12,569.141$

$B = -0.917$

$\therefore y = 12,569.141e^{-0.917x}$

3)

x	0	2	4
y	5.02	10	31.62

x	y	$\ln y$	x^2	$x \ln y$
0	5.02	1.613	0	0
2	10	2.303	4	4.606
4	31.62	3.454	16	13.816
Σ 6		7.370	20	18.422

$3A + 6B = 7.37$

$6A + 20B = 18.422$

$A = 1.536 = \ln a \Rightarrow a = 4.65$

$B = 0.46$

$\therefore y = 4.65e^{0.46x}$

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Correlation and Regression

Correlation and correlation coefficient.

A correlation is a statistical measure of the relationship b/n 2 variables. When 2 sets of data are strongly linked together, we say that they have correlation.

Correlation is +ve when the values ↑(↓) together, i.e, when the values of 1 variable increase (decrease), as the value of other ↑(↓). This is known as +ve correlation.

A correlation is -ve when the value decreases as other increases i.e, when as the value of 1 variable ↓ (↑) as the value of another ↑(↓). To form an inverse relationship, this is known as negative correlation.

+ve correlation
- The more u exercise, the more calories burn.
- The more it rains, the more sales for umbrellas go up.

Eg for -ve correlation
- The more one works, the less free time one has
- If the speed of the train ↑, the length of the time to go to the final pt ↓.

The correlation coefficient is a value that indicates the strength of the relationship b/n the variables. The coefficient can take any value from -1 to +1.

The numerical measure of correlation b/n 2 variables x and y. is known as Pearson's coefficient of correlation. usually denoted by 'r' and is defined as follows:

$$r = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{n\,\sigma_x\,\sigma_y} \quad \text{— (1)}$$

Where, $\sigma_x \rightarrow$ ST.D deviation of x

$$\bar{x} = \frac{\Sigma x}{n} \text{ (Mean)} \qquad \bar{y} = \frac{\Sigma y}{n}$$

$$\sigma_x^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2$$

OR

$$\sigma_x^2 = \frac{\Sigma (x-\bar{x})^2}{n}$$

$$\sigma_y^2 = \frac{\Sigma y^2}{n} - (\bar{y})^2$$

$$\sigma_y^2 = \frac{\Sigma (y-\bar{y})^2}{n}$$

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If $X = x - \bar{x}$ and $Y = y - \bar{y}$

then $\sigma_x^2 = \frac{\Sigma X^2}{n}$ and $\sigma_y^2 = \frac{\Sigma Y^2}{n}$

$\Rightarrow \sigma_x = \sqrt{\frac{\Sigma X^2}{n}}$ and $\sigma_y = \sqrt{\frac{\Sigma Y^2}{n}}$

$\sigma_x \cdot \sigma_y = \sqrt{\frac{\Sigma X^2 \cdot \Sigma Y^2}{n^2}}$

$= \frac{\sqrt{\Sigma X^2}\sqrt{\Sigma Y^2}}{n}$

OR $n\sigma_x \cdot \sigma_y = \sqrt{\Sigma X^2}\sqrt{\Sigma Y^2}$

$\therefore$ ① $\Rightarrow$ $\boxed{r = \frac{\Sigma XY}{\sqrt{\Sigma X^2}\sqrt{\Sigma Y^2}}}$ $\quad \Sigma XY \le \Sigma X^2 \cdot \Sigma Y^2$

P.T. $r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$

Proof: We shall P.T. $\sigma_x^2 + \sigma_y^2 = \sigma_{x-y}^2 + 2r\sigma_x\sigma_y$

Let $z = x - y$ — ①

$\Sigma z = \Sigma x - \Sigma y$

Dividing throughout by n,

$\frac{\Sigma z}{n} = \frac{\Sigma x}{n} - \frac{\Sigma y}{n}$

$\bar{z} = \bar{x} - \bar{y}$ — ②

① − ② $\Rightarrow z - \bar{z} = (x - \bar{x}) - (y - \bar{y})$

Now squaring on both sides,

$(z - \bar{z})^2 = \{(x - \bar{x}) - (y - \bar{y})\}^2$

$= (x - \bar{x})^2 + (y - \bar{y})^2 - 2(x - \bar{x})(y - \bar{y})$

$\frac{\Sigma(z - \bar{z})^2}{n} = \frac{\Sigma(x - \bar{x})^2}{n} + \frac{\Sigma(y - \bar{y})^2}{n} - \frac{2\Sigma(x - \bar{x})(y - \bar{y})}{n}$ — ③

W.K.T. $\sigma_x^2 = \frac{\Sigma(x - \bar{x})^2}{n}$

$\sigma_z^2 = \sigma_x^2 + \sigma_y^2 - \frac{2\Sigma(x - \bar{x})(y - \bar{y})}{n}$ — ④

W.K.T. $r = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{n\sigma_x\sigma_y}$

So $r\sigma_x\sigma_y = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{n}$

$\sigma_z^2 = \sigma_x^2 + \sigma_y^2 - 2r\sigma_x\sigma_y$

OR $\sigma_x^2 + \sigma_y^2 = \sigma_{x-y}^2 + 2r\sigma_x\sigma_y$

So $r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$

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Regression: Regression is an estimation of 1 independent variable in terms of other. If x and y are correlated, the best fitting straight line in the least square sense gives reasonable a good relation b/n x & y. The best fitting straight line of the form $y = ax + b$ (x being an independent variable) is called the regression line of y on x.

i.e. $x = ay + b$ (y being ind. variable) is called the regression line of x on y.

Equations of Regression Lines:

1) Regression line of y on x.

$$(y - \bar{y}) = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

2) Regression line of x on y.

$$(x - \bar{x}) = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

Proof:- Let $y = ax + b$ be the eqn of regression line of y on x.

The associated normal eqns are

$$\Sigma y = a\Sigma x + nb \quad \text{—— (1)}$$

$$\Sigma xy = a\Sigma x^2 + b\Sigma x$$

Divide (1) by n, $\frac{\Sigma y}{n} = \frac{a\Sigma x}{n} + b$

$$\Rightarrow \bar{y} = a\bar{x} + b$$

This shows that the regression line passes through $(\bar{x}, \bar{y})$.

Also, W.K.T. the eqn of straight line ~~line~~ passing through (x_1, y_1) having slope 'm' is

$$(y - y_1) = m(x - x_1) \quad \text{—— (2)}$$

If $(x_1, y_1) = (\bar{x}, \bar{y})$ then the above eqn becomes,

$$(y - \bar{y}) = m(x - \bar{x}) \quad \text{—— (2)}$$

or $Y = mX$ where $Y = y - \bar{y}$, $X = x - \bar{x}$

The normal eqns for fitting $Y = mX$ are

$$\Sigma Y = m\Sigma X$$

$$\Sigma XY = m\Sigma X^2 \Rightarrow m = \frac{\Sigma XY}{\Sigma X^2} \quad \text{—— (3)}$$

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Since the coefficient of correlation,

$$r = \frac{\Sigma xy}{n\sigma_x\sigma_y} \Rightarrow \Sigma xy = nr\sigma_x\sigma_y$$

Also we have $\sigma_x^2 = \frac{\Sigma x^2}{n} \Rightarrow \Sigma x^2 = n\sigma_x^2$

$$② \Rightarrow m = \frac{nr\sigma_x\sigma_y}{n\sigma_x^2}$$

$$m = \frac{r\sigma_y}{\sigma_x}$$

Thus $① \Rightarrow \boxed{(y-\bar{y}) = \frac{r\sigma_y}{\sigma_x}(x-\bar{x})}$ is the regression line of y on x

To prove $(x-\bar{x}) = r\frac{\sigma_x}{\sigma_y}(y-\bar{y})$

Let $x = ay+b$ be the eqn of regression line of x on y.

The associated normal eqns are

$$\Sigma x = a\Sigma y + nb \quad \text{—— ①}$$

and $\Sigma xy = a\Sigma y^2 + b\Sigma y$

Divide ① by n, $\frac{\Sigma x}{n} = a\frac{\Sigma y}{n} + b$

$$\Rightarrow \bar{x} = a\bar{y} + b$$

This shows that the regression line passes through $(\bar{x}, \bar{y})$. Also W.K.T, the eqn of straight line passing through (x_1, y_1) having slope m is,

$$(x-x_1) = m(y-y_1)$$

If $(x_1, y_1) = (\bar{x}, \bar{y})$ then above eqn becomes,

$$(x-\bar{x}) = m(x-\bar{x}) \quad \text{——②}$$

or $X = mY$ where $X = x-\bar{x}$, $Y = y-\bar{y}$

The normal eqns for fitting $X=mY$ are

$$\Sigma X = m\Sigma Y$$

$$\Sigma XY = m\Sigma Y^2$$

$$\Rightarrow m = \frac{\Sigma XY}{\Sigma Y^2} \quad \text{——} ③$$

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Since the coefficient of correlation,

$$r = \frac{\Sigma XY}{n\sigma_x\sigma_y} \Rightarrow \Sigma XY = nr\sigma_x\sigma_y$$

Also we have $\sigma_y^2 = \frac{\Sigma Y^2}{n}$ or $\Sigma Y^2 = n\sigma_y^2$

③ $\Rightarrow$ $m = \frac{nr\sigma_x\sigma_y}{n\sigma_y^2}$

$$m = \frac{r\sigma_x}{\sigma_y}$$

$\therefore$ ② $\Rightarrow$ $\boxed{(x-\bar{x}) = \frac{r\sigma_x}{\sigma_y}(y-\bar{y})}$ is the eqn for regression line of x on y

Note: The regression line of y on x is $(y-\bar{y}) = b_{yx}(x-\bar{x})$ where $b_{yx} = r\frac{\sigma_y}{\sigma_x}$

& x on y is $(x-\bar{x}) = b_{xy}(y-\bar{y})$ where $b_{xy} = r\frac{\sigma_x}{\sigma_y}$

$$b_{xy}\cdot b_{yx} = \left(r\frac{\sigma_y}{\sigma_x}\right)\left(r\cdot\frac{\sigma_x}{\sigma_y}\right)$$

$$\therefore r^2 = b_{xy}\cdot b_{yx}$$

$$r = \pm\sqrt{b_{xy}\cdot b_{yx}}$$

$$r = \pm\sqrt{(\text{coefficient of } x)(\text{coefficient of } y)} \rightarrow \text{GM}$$

This says r is the geometric mean of the regression coefficients.

Case(i): If b_{yx} is +ve and b_{xy} is +ve, then r is +ve

(ii): If b_{yx} is -ve and b_{xy} is -ve, then r is -ve

Angle bn regression lines

* If θ is the angle bn the lines of regression. Then prove that

$$\tan\theta = \frac{\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}\left(\frac{1-r^2}{r}\right).$$

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W.K.T. the lines of regression are

$$(y-\bar{y}) = r\frac{\sigma_y}{\sigma_x}(x-\bar{x}) \quad \& \quad (x-\bar{x}) = r\frac{\sigma_x}{\sigma_y}(y-\bar{y})$$

$$\text{or } (y-\bar{y}) = \frac{1}{r}\frac{\sigma_y}{\sigma_x}(x-\bar{x}) \quad \text{— (2)}$$

(1)

Now the slope of eqn (1) and (2) are

$$m_1 = r\frac{\sigma_y}{\sigma_x} \qquad m_2 = \frac{1}{r}\frac{\sigma_y}{\sigma_x}$$

We know in general,

$$\tan\theta = \frac{m_2 - m_1}{1 + m_1 m_2}$$

i.e., $$\tan\theta = \frac{\frac{1}{r}\frac{\sigma_y}{\sigma_x} - r\frac{\sigma_y}{\sigma_x}}{1 + \left(r\frac{\sigma_y}{\sigma_x}\right)\left(\frac{1}{r}\frac{\sigma_y}{\sigma_x}\right)}$$

$$= \frac{\frac{\sigma_y}{\sigma_x}\left[\frac{1}{r} - r\right]}{1 + \frac{\sigma_y^2}{\sigma_x^2}}$$

$$= \frac{\frac{\sigma_y}{\sigma_x}\left[\frac{1-r^2}{r}\right]}{\frac{\sigma_x^2 + \sigma_y^2}{\sigma_x^2}}$$

$$\boxed{\tan\theta = \frac{\sigma_x \cdot \sigma_y}{\sigma_x^2 + \sigma_y^2}\left[\frac{1-r^2}{r}\right]}$$

Note:

1) If $r = \pm 1$ then $\tan\theta = 0$. This implies that 2 regression lines coincides. Hence the variables are perfectly correlated.

2) If $r = 0$, then $\tan\theta = \infty$. This implies that the 2 regression lines are $\perp$ar. Hence the variables are uncorrelated.

* The regression line of y on x is

$$(y-\bar{y}) = b_{yx}(x-\bar{x}) \quad \text{where } b_{yx} = r\frac{\sigma_y}{\sigma_x}$$

(1)

Since $r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \cdot \Sigma y^2}}$ and $\sigma_x = \sqrt{\frac{\Sigma x^2}{n}}$ and $\sigma_y = \sqrt{\frac{\Sigma y^2}{n}}$

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$$b_{yx} = r\frac{\sigma_y}{\sigma_x} = \frac{\Sigma XY}{\sqrt{\Sigma X^2}\sqrt{\Sigma Y^2}} \cdot \frac{\sqrt{\Sigma Y^2}}{\sqrt{n}} \cdot \frac{\sqrt{n}}{\sqrt{\Sigma X^2}}$$

$$\therefore \boxed{b_{yx} = \frac{\Sigma XY}{\Sigma X^2}}$$

Thus (1) $\Rightarrow$ $\boxed{Y = \frac{\Sigma XY}{\Sigma X^2} \cdot X}$

IIly the regression line of x on y is

$$(x - \bar{x}) = r\frac{\sigma_x}{\sigma_y}(y - \bar{y}) \Rightarrow X = \frac{\Sigma XY}{\Sigma Y^2} \cdot Y$$

Problem: Note (or)

(1) To find the coefficient of correlation and hence the regression lines. We use the following procedure:

x	y	$z = x - y$	x^2	y^2	z^2

$$\bar{x} = \frac{\Sigma x}{n} \quad ; \quad \bar{y} = \frac{\Sigma y}{n} \quad ; \quad \bar{z} = \frac{\Sigma z}{n}$$

$$\sigma_x^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2 \quad ; \quad \sigma_y^2 = \frac{\Sigma y^2}{n} - (\bar{y})^2 \quad ; \quad \sigma_z^2 = \frac{\Sigma z^2}{n} - (\bar{z})^2$$

$\therefore$ The coefficient of correlation, $r = \dfrac{\sigma_x^2 + \sigma_y^2 - \sigma_z^2}{2\sigma_x\sigma_y}$

Hence the lines of regression are $(y - \bar{y}) = r\dfrac{\sigma_y}{\sigma_x}(x - \bar{x})$

and $(x - \bar{x}) = r\dfrac{\sigma_x}{\sigma_y}(y - \bar{y})$

(2) To find the regression lines and hence the coefficient of correlation we use the following procedure.

$$\bar{x} = \frac{\Sigma x}{n} \text{ and } \bar{y} = \frac{\Sigma y}{n}$$

x	y	$X = x - \bar{x}$	$Y = y - \bar{y}$	X^2	Y^2

The lines of regression are

$(y - \bar{y}) = b_{yx}(x - \bar{x})$ where $b_{yx} = \dfrac{\Sigma XY}{\Sigma X^2}$

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$(x - \bar{x}) = b_{xy}(y - \bar{y})$ where $b_{xy} = \frac{\Sigma xy}{\Sigma y^2}$

Hence $r = \pm \sqrt{b_{yx} \cdot b_{xy}}$

$= \pm \sqrt{(\text{coefficient of } x) \cdot (\text{coefficient of } y)}$

1) Compute the coefficient of correlation and hence the lines of regression for the following data:

a)

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

b)

x	1	2	3	4	5
y	2	5	3	8	7

c)

x	1	2	3	4	5	6	7	8	9	10
y	10	12	16	28	25	36	41	49	40	50

Soln:-

x	y	$z = x - y$	x^2	y^2	z^2
1	9	-8	1	81	64
2	8	-6	4	64	36
3	10	-7	9	100	49
4	12	-8	16	144	64
5	11	-6	25	121	36
6	13	-7	36	169	49
7	14	-7	49	196	49
Σ 28	77	-49	140	875	347

$n = 7$

$\bar{x} = \frac{\Sigma x}{n} = \frac{28}{7} = 4$

$\bar{y} = \frac{\Sigma y}{n} = \frac{77}{7} = 11$

$\bar{z} = \frac{\Sigma z}{n} = \frac{-49}{7} = -7$

$\sigma_x^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2$

$= \frac{140}{7} - (4)^2$

$= 20 - 16$

$= 4$

$\sigma_y^2 = \frac{\Sigma y^2}{n} - (\bar{y})^2$

$= \frac{875}{7} - (11)^2$

$= 125 - 121$

$= 4$

$\sigma_z^2 = \frac{\Sigma z^2}{n} - (\bar{z})^2$

$= \frac{347}{7} - (-7)^2$

$= 49.57 - (49)$

$= 0.57$

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∴ coefficient of correlation, $r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_z^2}{2\sigma_x\sigma_y}$

$= \frac{4+4-0.57}{2(2)(2)} = \frac{7.43}{8} = 0.929$

$= 0.928 \simeq 0.93$

Hence the lines of regression are $(y-\bar{y}) = r\frac{\sigma_y}{\sigma_x}(x-\bar{x})$

and $(x-\bar{x}) = r\frac{\sigma_x}{\sigma_y}(y-\bar{y})$

i.e, $(y-11) = (0.93)\left(\frac{2}{2}\right)(x-4)$ and $(x-4) = 0.93\left(\frac{2}{2}\right)(y-11)$

or $(y-11) = 0.93(x-4)$ and $(x-4) = 0.93(y-11)$

$(y-11) = 0.93x - 3.72$; $x-4 = 0.93y - 10.23$

$y = 0.93x + 7.28$; $x = 0.93y - 6.23$

---

b) $n = 5$

x	y	$z = x-y$	x^2	y^2	z^2
1	2	−1	1	4	1
2	5	−3	4	25	9
3	3	0	9	9	0
4	8	−4	16	64	16
5	7	−2	25	49	4
Σ 15	25	−10	55	151	30

$\bar{x} = \frac{\Sigma x}{n} = \frac{15}{5} = 3$

$\bar{y} = \frac{\Sigma y}{n} = \frac{25}{5} = 5$

$\bar{z} = \frac{\Sigma z}{n} = \frac{-10}{5} = -2$

$\sigma_x^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2$

$= \frac{55}{5} - (3)^2$

$= 11 - 9$

$= 2$

$\sigma_y^2 = \frac{\Sigma y^2}{n} - (\bar{y})^2$

$= \frac{151}{5} - (5)^2$

$= 5.2$

$\sigma_z^2 = \frac{\Sigma z^2}{n} - (\bar{z})^2$

$= \frac{30}{5} - (-2)^2$

$= 6 - 4$

$= 2$

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coefficient of relation, $r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_z^2}{2\sigma_x\sigma_y}$

$= \frac{2 + 5.2 - 2}{2(\sqrt{2})(\sqrt{5.2})} = \frac{\sqrt{5.2}}{2\sqrt{2}} = \frac{0.8}{0.81}$

$\sqrt{5.2} = 2.28$

Hence the lines of regression are,

$(y - \bar{y}) = r\frac{\sigma_y}{\sigma_x}(x - \bar{x})$; $(x - \bar{x}) = r\frac{\sigma_x}{\sigma_y}(y - \bar{y})$

$(y - 5) = 0.8\left(\frac{2.28}{1.414}\right)(x - 3)$; $(x - 3) = (0.8)\left(\frac{1.414}{2.28}\right)(y - 5)$

$y - 5 = 1.289(x - 3)$; $x - 3 = 0.49(y - 5)$

$y - 5 = 1.289x - 3.867$; $x - 3 = 0.49y - 2.45$

$y = 1.289x + 1.133$; $x = 0.49y - 0.55$

$y = 1.306x + 1.052$ $x = 0.502y + 0.49$

2) Compute the lines of regression and hence the coefficient of correlation for the following data:

a)

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

b)

x	1	3	4	2	8	9	10	13	15
y	8	6	10	8	12	16	10	32	32

c)

x	17	18	19	19	20	20	21	22	23	23
y	12	16	14	11	15	19	22	15	16	20

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2) a)

x	y	$X = x - \bar{x}$	$Y = y - \bar{y}$	X^2	Y^2	XY
1	9	-3	-2	9	4	6
2	8	-2	-3	4	9	6
3	10	-1	-1	1	1	1
4	12	0	1	0	1	0
5	11	1	0	1	0	0
6	13	2	2	4	4	4
7	14	3	3	9	9	9
Σ 28	77			28	28	26

$\bar{x} = \frac{\Sigma x}{n} = \frac{28}{7} = 4$

$\bar{y} = \frac{77}{7} = 11$

$X = \frac{\Sigma XY}{\Sigma Y^2} \cdot Y$ $\quad Y = \frac{\Sigma XY}{\Sigma X^2} \cdot X$

or $(x - \bar{x}) = \frac{\Sigma XY}{\Sigma Y^2} \cdot (y - \bar{y})$ $\quad (y - \bar{y}) = \frac{\Sigma XY}{\Sigma X^2} \cdot (x - \bar{x})$

$(x - 4) = \frac{26}{28}(y - 11)$; $(y - 11) = \frac{26}{28}(x - 4)$

$x - 4 = 0.93y - 10.23$ $\quad y - 11 = 0.93x - 3.72$

$x = 0.93y - 6.23$ $\quad y = 0.93x + 7.28$

$y = 0.93x + 7.28$

$r = \pm \sqrt{\text{coefficient of } x \text{ and } y}$

$= \pm \sqrt{(0.93)(0.93)} = +0.93$

---

d)

x	10	14	18	22	26	30
y	18	12	24	6	30	36

Here $n = 6$ $\quad \bar{x} = \frac{\Sigma x}{n} = \frac{120}{6} = 20$

$\bar{y} = \frac{\Sigma y}{n} = \frac{126}{6} = 21$

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x	y	$X = x - \bar{x}$	$Y = y - \bar{y}$	X^2	Y^2	XY
10	18	-10	-3	100	9	30
14	12	-6	-9	36	81	54
18	24	-2	3	4	9	-6
22	6	2	-15	4	225	-30
26	30	6	9	36	81	54
30	36	10	15	100	225	150
Σ 120	126			280	630	252

The eqns of lines of regression are,

$$Y = b_{yx} X \quad \text{and} \quad X = b_{xy} Y$$

where $b_{yx} = \frac{\Sigma XY}{\Sigma X^2} = \frac{252}{280} = 0.9$

and $b_{xy} = \frac{\Sigma XY}{\Sigma Y^2} = \frac{252}{630} = 0.4$

$\therefore \quad Y = 0.9X \quad \text{and} \quad X = 0.4Y$

$(y - \bar{y}) = 0.9(x - \bar{x}) \quad \text{and} \quad (x - \bar{x}) = 0.4(y - \bar{y})$

$y - 21 = 0.9(x - 20) \quad \text{and} \quad x - 20 = 0.4(y - 21)$

$y - 21 = 0.9x - 18 \quad , \quad x - 20 = 0.4y - 8.4$

$y = 0.9x + 3 \quad \text{and} \quad x = 0.4y + 11.6$

Hence the coefficient of regression, $r = \pm \sqrt{b_{xy} \cdot b_{yx}}$

$r = \pm \sqrt{(0.9)(0.4)}$

$r = \pm 0.6$

Since b_{xy} and b_{yx} are +ve so, $r = +0.6$.

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1) $8x - 10y + 66 = 0$ and $40x - 18y = 214$ are the 2 regression lines. Find the mean value of x and y and the coefficient of correlation. Also find σ_y if $\sigma_x = 3$.

Soln:- since the regression lines passes through $(\bar{x}, \bar{y})$ we have

$8\bar{x} - 10\bar{y} = -66$ and $40\bar{x} - 18\bar{y} = 214$

$\bar{x} = 13$

& $\bar{y} = 17$

$8x - 10y = -66$

$\Rightarrow y = 0.8x + 6.6$ is regr. line of y on x

$40x - 18y = 214$

$40x = 18y + 214$

$\Rightarrow x = 0.45y + 5.35$ x on y.

Then $r = \pm\sqrt{(\text{coefficient of } x)(\text{coefficient of } y)}$

$= \pm\sqrt{(0.8)(0.45)} = \pm\sqrt{0.36}$

$= \pm 0.6$

since both the coefficients are +ve, then $r = +0.6$.

W.K.T. the coefficient of x, $b_{yx} = r \cdot \frac{\sigma_y}{\sigma_x}$

$0.8 = 0.6 \frac{(\sigma_y)}{3}$

$\sigma_y = 4$

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2) Given the eqn of regression lines $x = 19.13 - 0.87y$ and $y = 11.64 - 0.5x$. Compute the mean value of x, y and r.

Soln:-

Coefficient of correlation, $r = \pm\sqrt{(\text{coefficient of } x)(\text{coefficient of } y)}$

$= \pm\sqrt{(-0.5)(-0.87)}$

$= \pm 0.66$

since both are negative, $r = -0.66$

since lines pass through $(\bar{x}, \bar{y})$

$\bar{x} = 19.13 - 0.87\bar{y}$ or $\bar{x} + 0.87\bar{y} = 19.13$ $\quad \bar{x} = 15.93$

$\bar{y} = 11.64 - 0.5\bar{x}$ $\quad 0.5\bar{x} + \bar{y} = 11.64$ $\quad \bar{y} = 3.67$

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3) Compute from the following regression lines.

$\bar{x}, \bar{y}$ and r

i) $2x + 3y + 1 = 0$

$x + 6y - 4 = 0$

$x = -1.5y - 0.5$

$y = -0.167x + 1/6$

$\bar{x} + 1.5\bar{y} + 0.5 = 0$ ii)

$-0.167\bar{x} - \bar{y} + 1/6 = 0$

$\bar{x} = -2$

$\bar{y} = 1$

$r = 0.5$

$y = x$ $\quad x - y = 0$

$4x - y - 3 = 0$

$\bar{x} = 1$

$\bar{y} = 1$

$x = y$

$y = 4x - 3$

$r = \sqrt{\text{product}}$

$= 2 \times$

so consider other

$y = x$

$x = 0.25y + 0.75$

now $r = \sqrt{0.25} = 0.5$

4) If the correlation coefficient bln 2 variables x and y is 0.5 and the acute angle bln lines of regression is $\tan^{-1}\left(\frac{3}{5}\right)$, S.T.

$\sigma_y = 2\sigma_x$ or $\sigma_x = 2\sigma_y$.

Soln: Given $r = 0.5$

$\theta = \tan^{-1}\left(\frac{3}{5}\right)$

We have $\tan\theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}\left[\frac{1 - r^2}{r}\right]$

$\tan\left(\tan^{-1}\left(\frac{3}{5}\right)\right) = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}\left[\frac{1 - (0.5)^2}{0.5}\right]$

$\frac{3}{5} = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}\left[\frac{1 - \frac{1}{4}}{1/2}\right]$ $\quad \frac{\frac{3}{4}}{\frac{1}{2}}$

$\frac{3}{5} = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}\left[\frac{3}{2}\right]$

$\frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} = \frac{2}{5}$

or $2\sigma_x^2 + 2\sigma_y^2 - 5\sigma_x\sigma_y = 0$

$2\sigma_x^2 + 5\sigma_x\sigma_y + 2\sigma_y^2 = 0$

$2\sigma_x^2 - 4\sigma_x\sigma_y - \sigma_x\sigma_y + 2\sigma_y^2 = 0$

$2\sigma_x[\sigma_x - 2\sigma_y] - \sigma_y[\sigma_x - 2\sigma_y] = 0$

or $\sigma_x - 2\sigma_y = 0$ and or $2\sigma_x = \sigma_y$

so $\boxed{\sigma_x = 2\sigma_y}$ or $\boxed{\sigma_y = 2\sigma_x}$

5) In a bivariate distribution, $\sigma_x = \sigma_y$ and the angle bln the regression lines is $\tan^{-1}(3)$. Find r.

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Soln:- $\tan\theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}\left[\frac{1-r^2}{r}\right]$ $\quad \sigma_x = \sigma_y$

$$\tan(\tan^{-1}3) = \frac{\sigma_x \cdot \sigma_y}{\sigma_x^2 + \sigma_x^2}\left[\frac{1-r^2}{r}\right]$$

$$3 = \frac{\sigma_x^2}{2\sigma_x^2}\left[\frac{1-r^2}{r}\right]$$

$$6 = \frac{1-r^2}{r} \Rightarrow 6r = 1-r^2 \Rightarrow r^2 + 6r - 1 = 0$$

$$r = \frac{-6 \pm \sqrt{36 - 4(1)(-1)}}{2(1)} = \frac{-6 \pm \sqrt{40}}{2}$$

$$r = \frac{-6+\sqrt{40}}{2} \quad \text{or} \quad r = \frac{-6-\sqrt{40}}{2}$$

$r = +0.16$ $\qquad r = -6.16$ X $\qquad -1 \le r \le 1$

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